FARM MANAGEMENT SYSTEM

User Stories

In the Farm Management System (FMS), as a farmer, farm manager, or system administrator, I want to use a centralized Salesforce-based platform to record and manage farm, crop, harvest, and lower information so that all operational data is stored securely and systematically. I want to enter and update farm details, crop production records, and lower transactions easily through a user-friendly interface with proper validation rules to prevent incorrect entries. I also want the system to generate real-time reports and interactive dashboards so that I can analyze farm productivity, monitor sales, and make informed decisions. Additionally, I want role-based access and secure data storage to ensure that sensitive information is protected while enabling smooth collaboration among users. This system will help reduce manual paperwork, minimize errors, improve efficiency, and support data-driven farm management.

1.INTRODUCTION

1.1 Project Overview:

"The Farm Management System (FMS) is a Salesforce-based application designed to digitize and streamline agricultural operations. It manages data related to farms, crops, farmers, and buyers while providing real-time insights through reports and dashboards. The system helps farmers and managers monitor agricultural activities efficiently and make informed decisions."

1.2 Purpose:

The main purpose of this project is to:

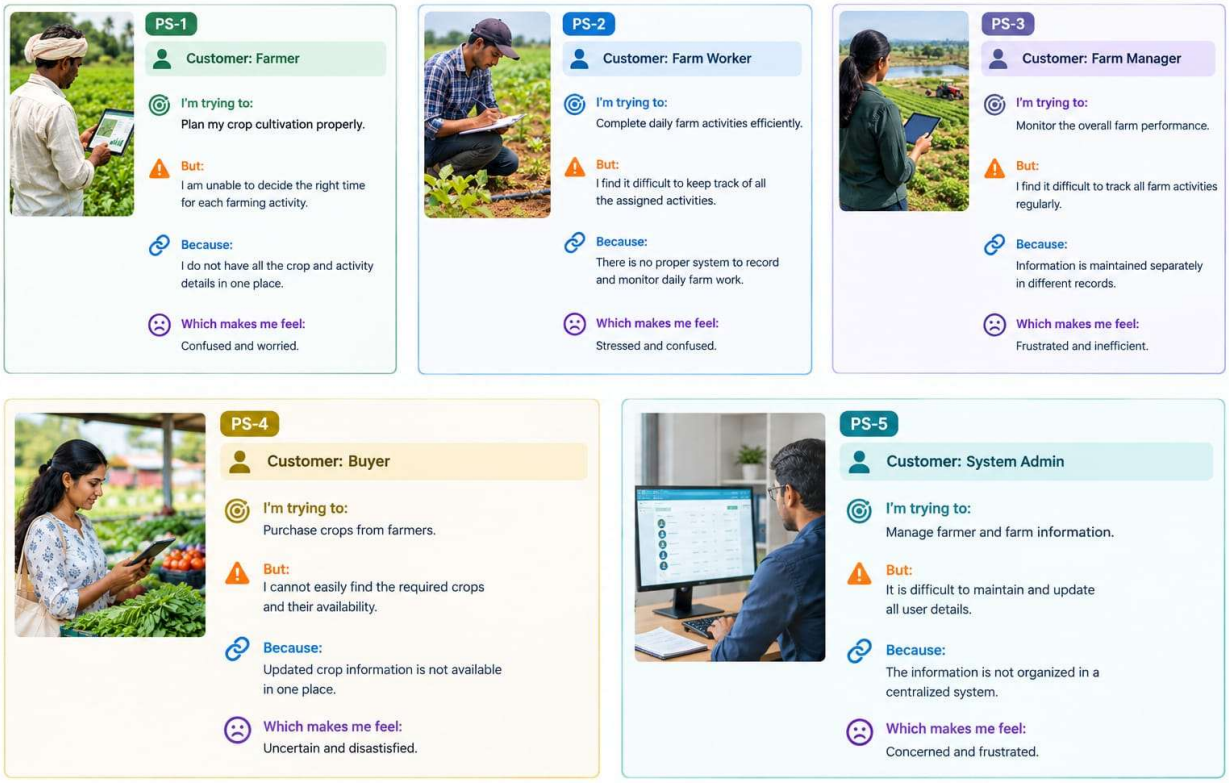
- Digitize agricultural data management
- Improve data accuracy and accessibility
- Automate record handling
- Enable real-time monitoring
- Support data-driven farming decisions
- Enhance productivity and transparency

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Farmer	Plan my crop cultivation properly	I am unable to decide the right time for each farming activity	I do not have all the crop and activity details in one place	Confused and Worried
PS-2	Farm Worker	Complete	I am unable	Tasks are	Stressed

		my daily field work on time	to track my assigned tasks properly	given verbally and there is no system to record daily activities	and unorganized
PS-3	Farm Manager	Monitor the overall farm performanc e	I find it difficult to track all farm activities regularly	Information is maintained separately in different records	Frustrated and inefficient
PS-4	Buyer	Purchase crops from farmers directly	I cannot easily find the required crops and their availability	There is no centralized system to show live crop stock	Uncertain and dissatisfied
PS-5	System Admin	Manage farmer and farm information	It is difficult to maintain and update all user details	The information is not organized in a centralized system	Concerned and frustrated



2.2 Empathy Map Canvas

Target users:

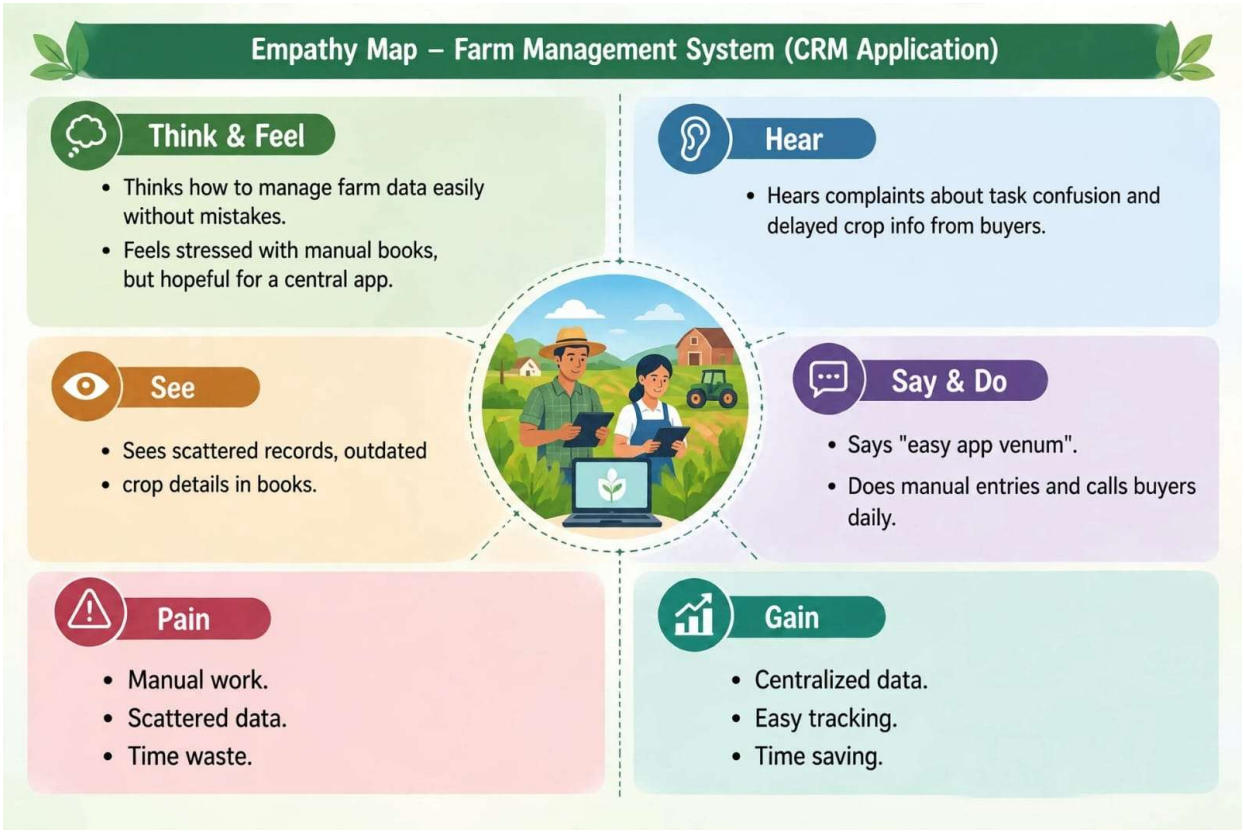
- Farmers
- Farm Workers
- Farm Managers
- Buyers
- System Administrators

Empathy Analysis:

- **Think & Feel:** Thinks how to manage farm data easily without mistakes.Feels stressed with manual books,but hopeful for a central app.

- **Hear :** Hears complaints about task confusion and delayed crop info from buyers.

- **See :** Sees scattered records,outdated crop details in books.
- **Say & Go :** Says “easy app venum”.Does manual entires and calls buyers easily.
- **Pain :** Manual work,scattered data,time waste.
- **Gain :** Centralized data, easy tracking,time saving.



2.3 Brainstorming

Brainstormed ideas

- Centralized Farmer Database System
- Smart Crop Activity & Task Assignment

- Live Crop Availability Dashboard for Buyers
- Farm Expense & Profit Tracker
- Role-Based SalesForce Farm Management System

Selected idea

Role-Based Salesforce Farm Management System



3.REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages :

1. Secure Role-Based Login

2. Centralized Dashboard Access

3. Farm & Crop Data Entry

4. Farmer & Buyer Management

5. Task Assignment for Workers

6. Expense & Profit Tracker

7. Reports for Decision Making



3.2 Solution Requirements

Functional Requirements

FR	Requirement
FR-1	User Login & Authentication
FR-2	Dashboard Management
FR-3	Farm & Crop Data Management
FR-4	Farmer & Buyer Management
FR-5	Task Assignment & Tracking
FR-6	Expense Tracker
FR-7	Reports & Analytics

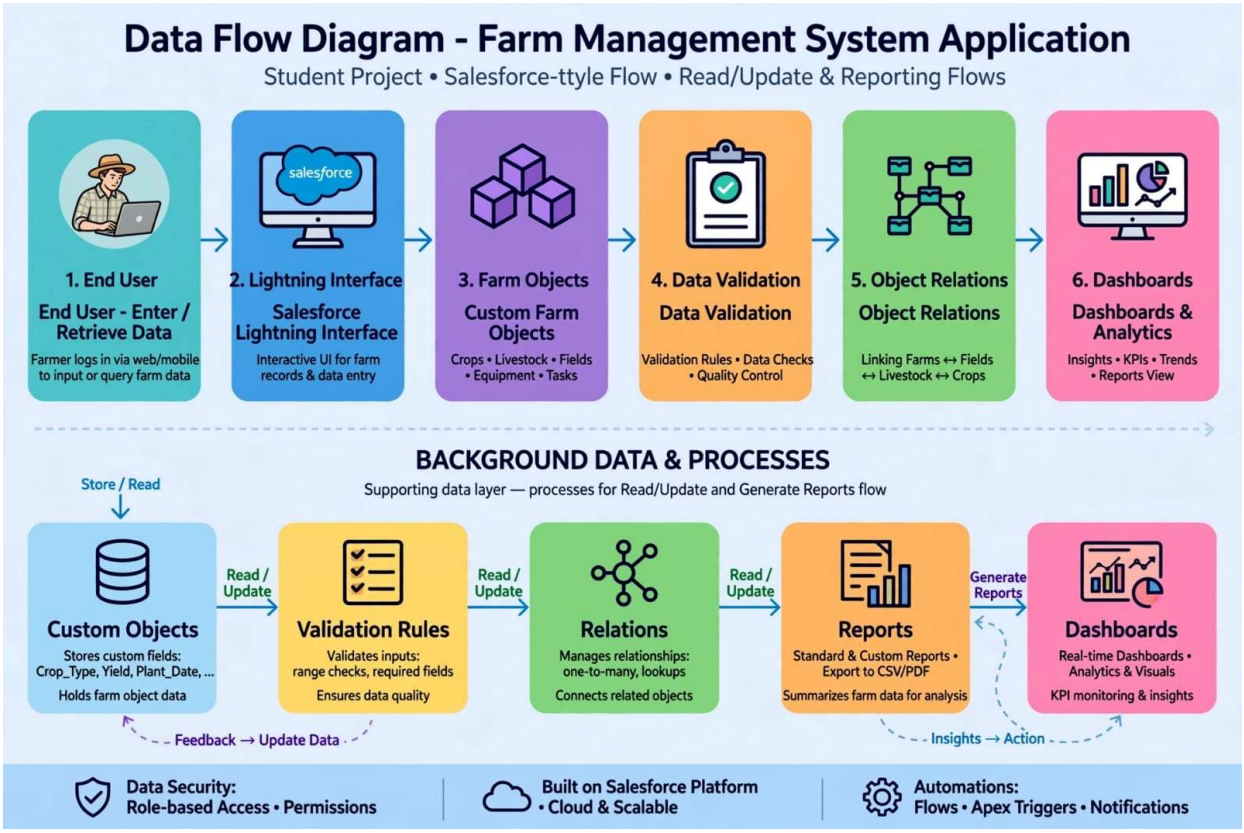
Non-Functional Requirements

NFR	Description	Example
Usability	System should have simple and user-friendly for farmers with less technical knowledge	Easy navigation with icons and regional language support
security	System should protect farm,crop and buyer data farm unauthorized access	Role-based access-only Admin can delete data
Reliability	System should store all data correctly without any loss	Auto backup even if internet fails
Performance	System should load dashboard and reports quickly	Reports should load within 3 seconds
Availability	System should be available anytime for farmers	24/7 access through cloud platform
Scalability	System should handle more farms,users and data in future	Supports 100+ farmers without slow performance

3.3 Data Flow Diagram

Data Flow

End User -> Authentication -> Lightning UI -> Custom -> Farm Objects -> Validation Checks -> Inter-Objects Relationships -> Reports -> Dashboard Visualization



3.4 Technology Stack

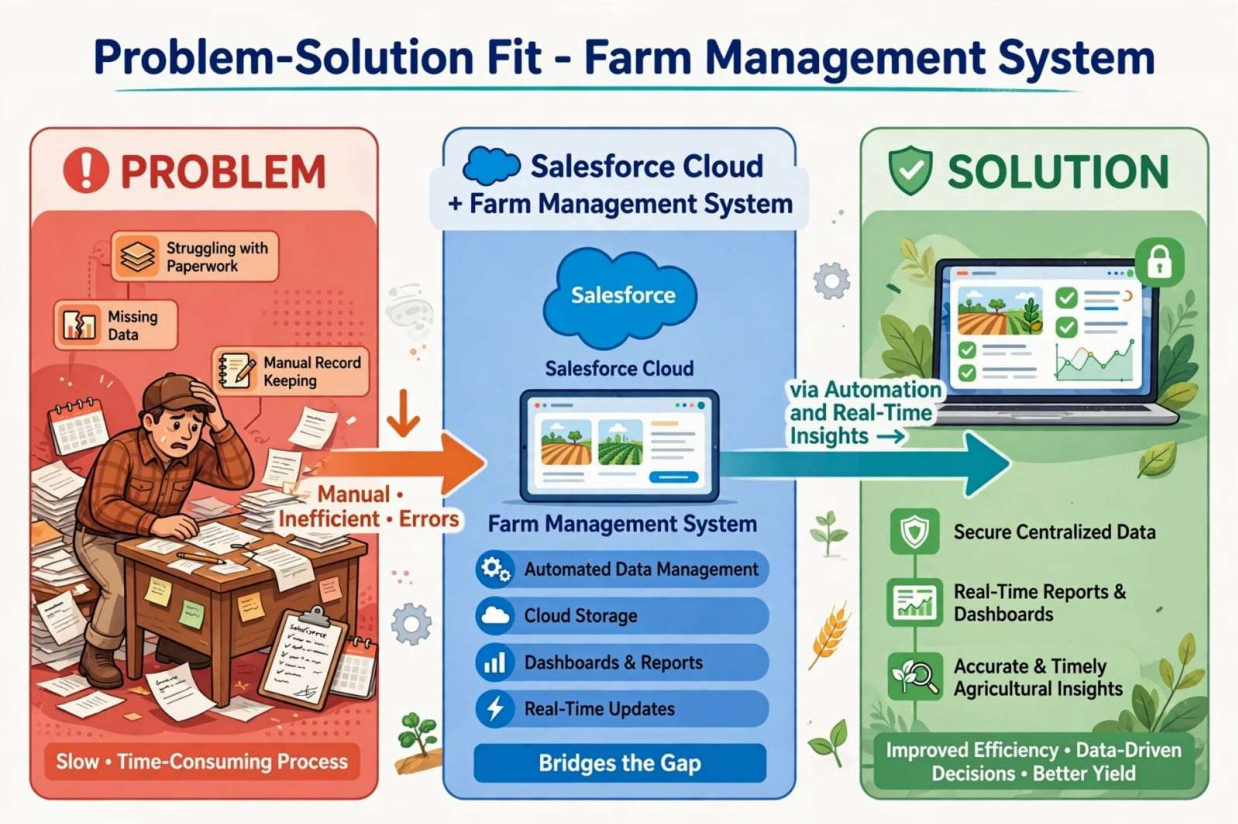
Layer	Technology
Frontend	Salesforce Lightning UI with Custom Components

Backend	Salesforce Platform (Apex Classes,Triggers)
Database	Salesforce cloud storage with Custom objects
Automation	Validation Rules,Flows and Process Automation
Analytics	Dynamic Reports & Interactive Dashboards

4.PROJECT DESIGN PHASE

4.1 Problem-Solution Fit

Farmers currently depend on manual notes and scattered Excel sheets for crop tracking, which causes missing data and delayed decisions. Our Farm Management System built on Salesforce converts this manual work into a centralized cloud system. With automated workflows,secure data storage,and live dashboards,farmers get accurate crop and buyer datails instantly without data loss or time delay .



4.2 Proposed Solution

Functional Requirements

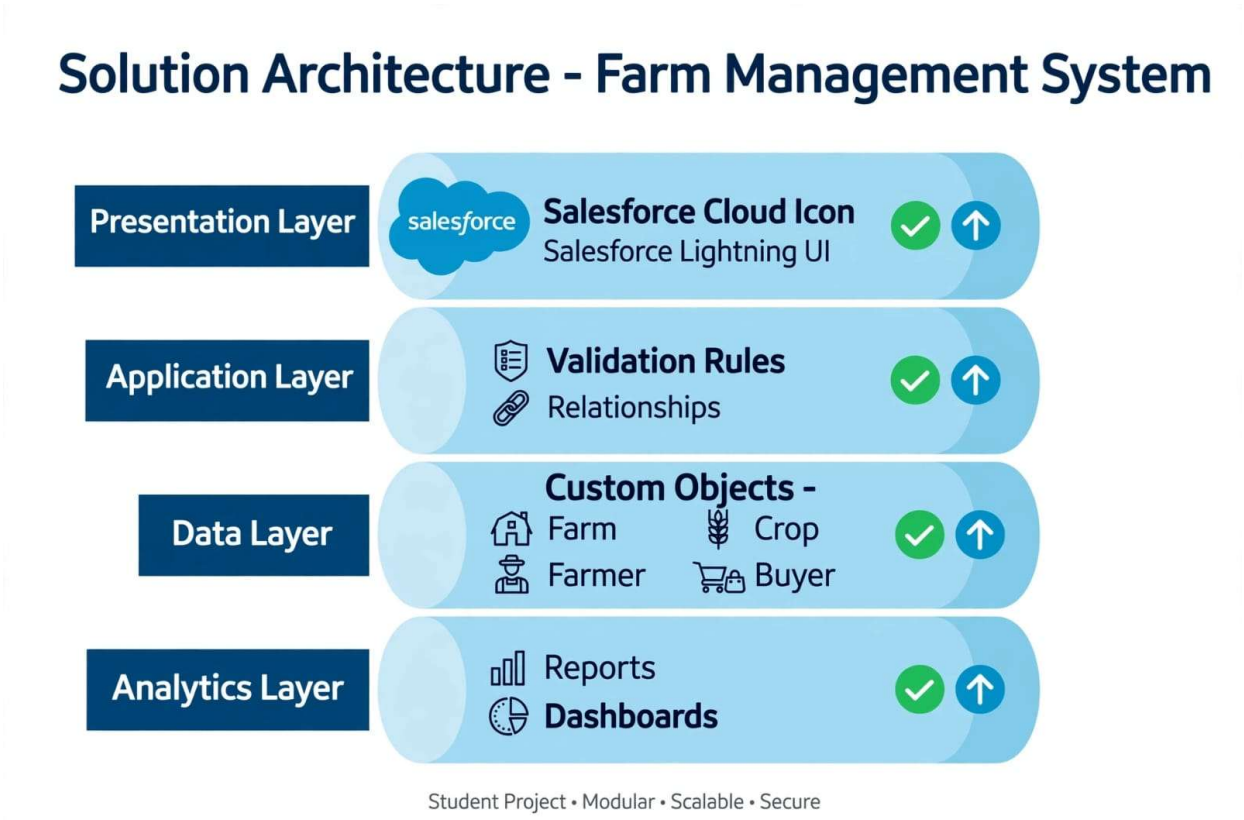
Parameter	Description
Problem	Farmers currently store farm,crop,farmer and buyer details in notebooks and Excel sheets,which creates errors,missing data and reporting delays.
Solution	Developing a centralized Farm Management System on Salesforce Lighting Platform with automated data handling and secure cloud database.
Innovation	Includes smart data validation,

	auto-linking between Farm-Crop-Buyer objects,real-time cloud sync and instant dashboards analytics.
Impact	Increases data accuracy to 99%,reduces manual work time,provides faster crop reports and helps farmers take better decisions.
Business Model	A digital agriculture service platform offering farm data management ,crop tracking and buyer connection services.
Scalability	Designed to support unlimited farms, users and locations through Salesforce scalable cloud infrastructure.

4.3 Solution Architecture

Architecture Layers

1. **Presentation** - Interactive Salesforce Lightning Interface.
2. **Application Layer** - Business Logic with Validation Rules,Flows and Relationships.
3. **Data Layer** - Custom Objects Storage.
4. **Analytics Layer** - Smart Reports & Dashboards.



5.PROJECT PLANING & SCHEDULING

5.1 Project Planning

- Created Salesforce Developer Org
- Developed custom objects
- Configured tabs and Lightning App
- Created fields and relationships
- Designed profiles and roles
- Created users and sharing rules
- Developed reports and dashboards

● Conducted testing and reviews

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule.

Sprint	Functional Requirement (Epic)	User Story Number	User Story/Task	Story points	Priority	Team Member
Sprint-1	Developer setup	USN-1	As a system administrator, I want to create and configure a Salesforce developer account so that I can set up and deploy the Farm Management System.	3	High	Member2
Sprint-2a	Data Modeling	USN-2	As an admin, I want to create Farm, Crop, Farmer, and Buyer objects with proper relationship	5	High	Member2

			s so that agricultural data is stored and managed efficiently			
Sprint-2	Data Modeling	USN-3	As a staff member, I want tabs and a Lightning App for all farm modules so that I can easily navigate and perform daily operations.	5	High	Member1
Sprint-3	Automation	USN-4	As a system, I want to validate farm and crop details such as dates, quantities, and mandatory fields so that incorrect or incomplete records are prevented.	3	High	Member4
Sprint-3	Automation	USN-5	As a manager, I	3	High	Member4

			want automated flows for crop updates and buyer notifications so that farm operations are processed faster without manual intervention.			
Sprint-4	Security	USN-6	As an administrator, I want role-based access control for farmers, managers, and buyers so that sensitive farm and transaction Data is secure.	5	High	Member1
Sprint-5	Reports	USN-7	As a farm manager, I want reports showing crop yield, sales, and buyer transactions so that I	4	High	Member5

			can monitor farm performance and Make informed decisions.			
Sprint-6	Dashboards	USN-8	As a farm manager, I want dashboards showing crop productivity, sales trends, and operational statistics so that I can monitor performance visually.	4	High	Member3

Project Tracker,Velocity & Burndown Chart :

Sprint	**Total Story Points**	**Duration**	**Sprint Start Date**	**Sprint End Date(Planned)**	**Story Points Completed (as on Planned End Date)**	**Sprint Release Date (Actual)**
Sprint-1	20	2 Days	23 Sep 2026	24 Sep 2026	3	24 Sep 2026
Sprint-2	20	1 Days	25 Sep 2026	25 Sep 2026	5	25 Sep 2026

Sprint-3	20	2 Days	26 Sep 2026	27 Sep 2026	3	27 Sep 2026
Sprint-4	20	1 Days	28 Sep 2026	28 Sep 2026	4	28 Sep 2026

6.PROJECT DEVELOPMENT PHASE :

Project Description:

The Farm Management System (FMS) in Salesforce is a comprehensive solution designed to digitize and streamline agricultural operations. This application empowers farmers, farm managers, and stakeholders by managing data related to crops, Farm, Farmer, and buyers, while providing actionable insights through reports, and dashboards.

Milestone 1-Salesforce:

What is Salesforce?

Salesforce is your customer success platform, designed to help you sell, service, market, analyze, and connect with your customers. Salesforce has everything you need to run your business from anywhere. Using standard products and features, you can manage relationships with prospects and customers, collaborate and engage with employees and partners, and store your data securely in the cloud.

Activity1:

Creating a Salesforce Developer Org:

A Developer org has all the features and licenses you need to get started with Salesforce.

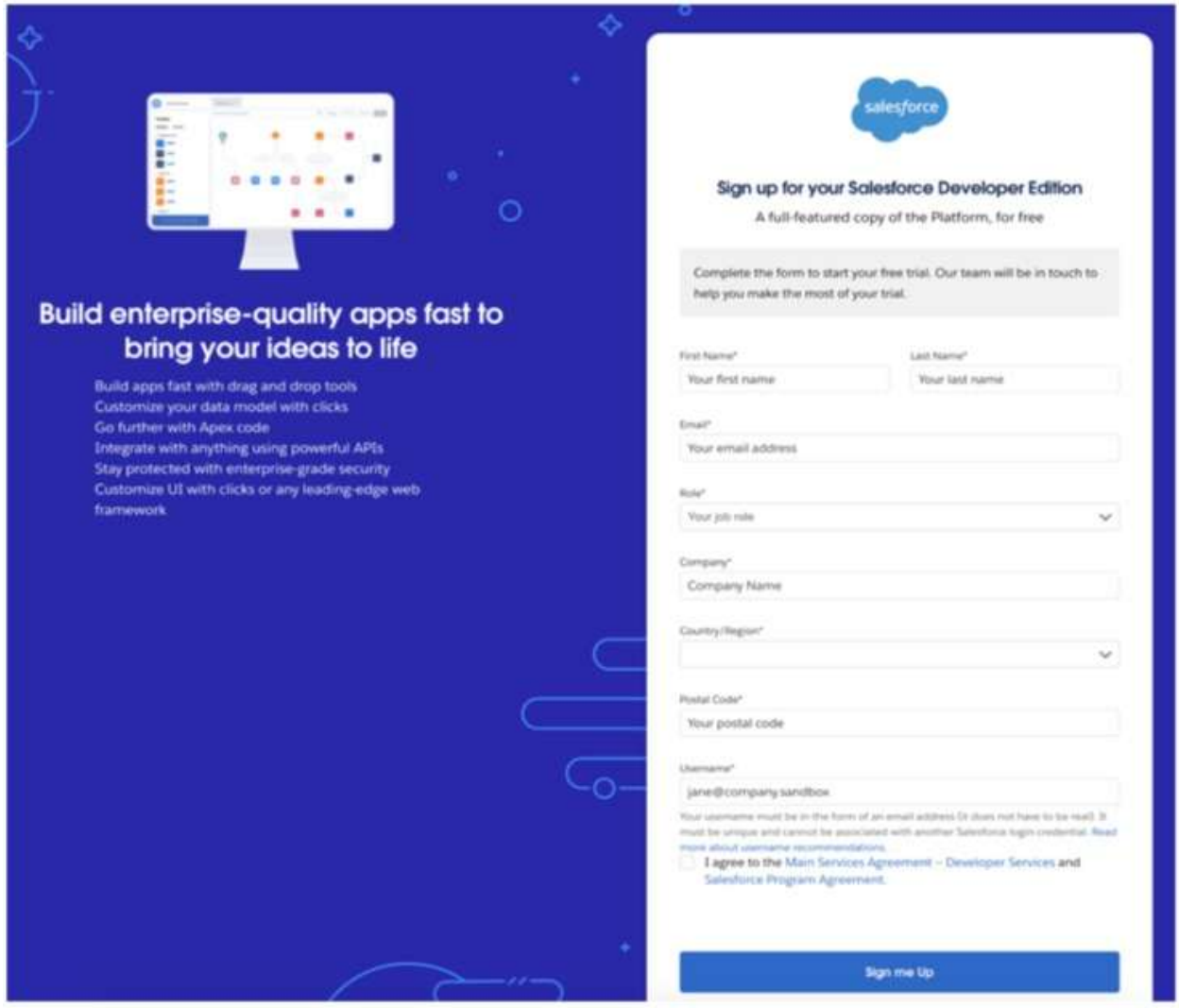
1. Search Developer.salesforce.com/signup .

2. Enter the following details like First name, last name, Email, Role, Company, Country/Region, Postal code, and Username must be unique.

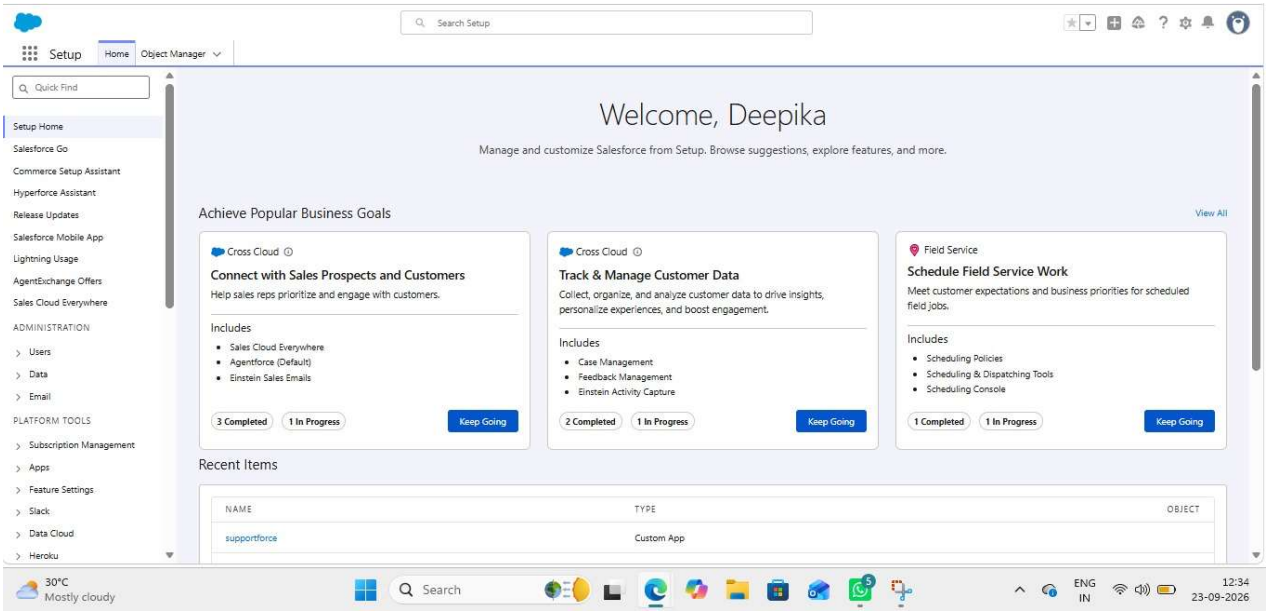
3. Click sign me up, after a few min you will reserve a mail salesforce org and by using the verify account link you can create your new password.

4. Click save.

5. Search login.salesforce.com



6..By using username and password you can into the Salesforce org.The homepage will appear below.


Search Setup
Setup
Home
Object Manager
Quick Find
Setup Home
Salesforce Go
Commerce Setup Assistant
Hyperforce Assistant
Release Updates
Salesforce Mobile App
Lightning Usage
AgentExchange Offers
Sales Cloud Everywhere
ADMINISTRATION
Users
Data
Email
PLATFORM TOOLS
Subscription Management
Apps
Feature Settings
Slack
Data Cloud
Heroku
Welcome, Deepika
Manage and customize Salesforce from Setup. Browse suggestions, explore features, and more.
Achieve Popular Business Goals
View All
Cross Cloud
Connect with Sales Prospects and Customers
Help sales reps prioritize and engage with customers.
Includes
Sales Cloud Everywhere
Agentforce (Default)
Einstein Sales Emails
3 Completed
1 In Progress
Keep Going
Cross Cloud
Track & Manage Customer Data
Collect, organize, and analyze customer data to drive insights, personalize experiences, and boost engagement.
Includes
Case Management
Feedback Management
Einstein Activity Capture
2 Completed
1 In Progress
Keep Going
Field Service
Schedule Field Service Work
Meet customer expectations and business priorities for scheduled field jobs.
Includes
Scheduling Policies
Scheduling & Dispatching Tools
Scheduling Console
1 Completed
1 In Progress
Keep Going
Recent Items
NAME
TYPE
OBJECT
supportforce
Custom App
30°C
Mostly cloudy
Search
ENG
IN
12:34
23-09-2026

Milestone 2-Object:

What is an object?

Salesforce objects are database tables that permit you to store data that is specific to an organization. It consists of fields (columns) and records (rows).

In This Application We Use 4 Custom Objects:

1. Farm
2. Crop
3. Farmer
4. Buyer

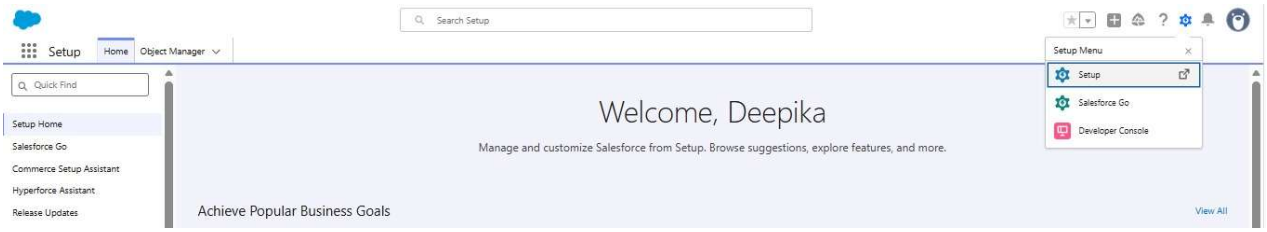
Activity 1 :

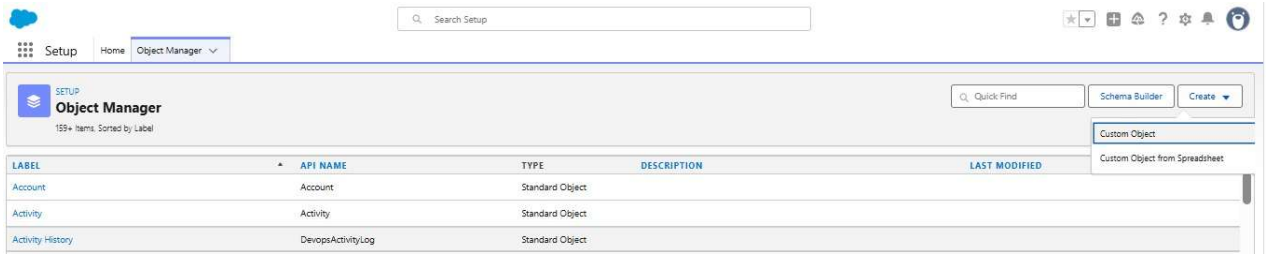
Create a custom object for Farm:

To create a custom object, follow these steps:

1. From setup click on object manager.

2. Click create, select custom object.





3. Fill in the label as " Farm ".

4. Fill in the plural label as " Farms".

5. Record name: Farm Name"

6. Select the data type as "Text.



7. In the Optional Features section, select Allow Reports and Track Field History.

8. In the Deployment Status section, ensure Deployed is selected
.
9. In the Search Status section, select Allow Search.

10.In the Object Creation Options section, select Add Notes and Attachments related list default page layout.

11. Leave everything else as is, and click Save.

Activity 2:

Creation of Crop object:
1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
a. On the Custom Object Definition page, create the object as follows:
b. Label: Crop
c. Plural Label: Crops
d. Record Name: Crop Name
e. Select the data type as "Text".
f. Check the Allow Reports checkbox g. Check the Allow Search checkbox
h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout
i. Click Save.

Activity 3:

Creation of Farmer object:
1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.

a. On the Custom Object Definition page, create the object as follows:
b. Label: Farmer.
c. Plural Label: Farmers
d. Record Name: Farmer Name
e. Select the data type as "Text".
f. Check the Allow Reports checkbox.
g. Check the Allow Search checkbox.
h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.
i. Click Save.

Activity 4:

Creation of Buyer object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click
on that and select Custom Object.

a. On the Custom Object Definition page, create the object as follows:
b. Label: Buyer
c. Plural Label: Buyers
d. Record Name: Buyer Name
e. Select the data type as "Text".
f. Check the Allow Reports checkbox.
g. Check the Allow Search checkbox.
h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.
i. Click Save.

Milestone 3:

Tab : Tab

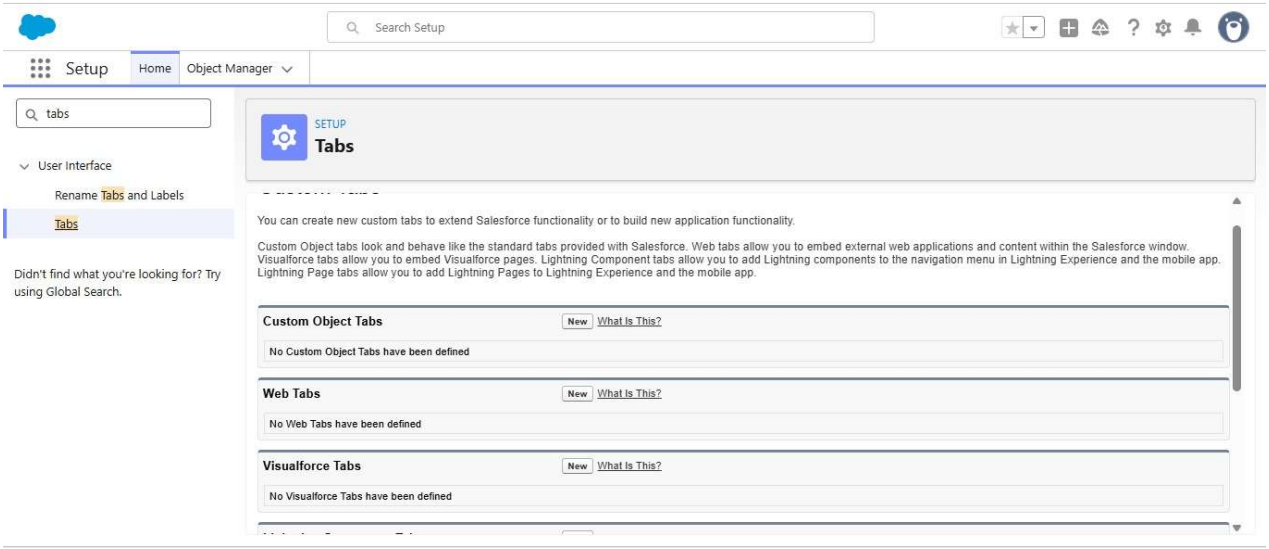
Tabs in Salesforce help users view the information at a glance. It displays the data of objects and other web content in the application.

Activity 1:

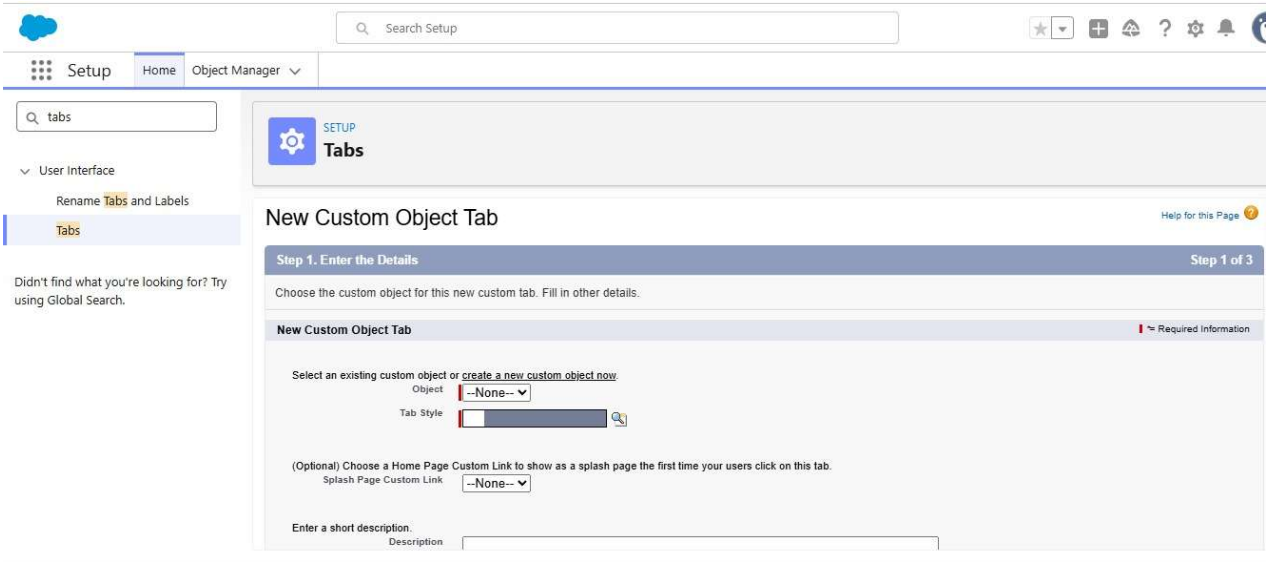
Creation of Farm tab:

Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.



2. Under custom object tabs, click New.



3. For Object, select Farm.
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 2:

Creation of Crop tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Crop.
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 3:

Creation of Farmer tab:

Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Farmer.
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 4:

Creation of Buyer tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Buyer
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

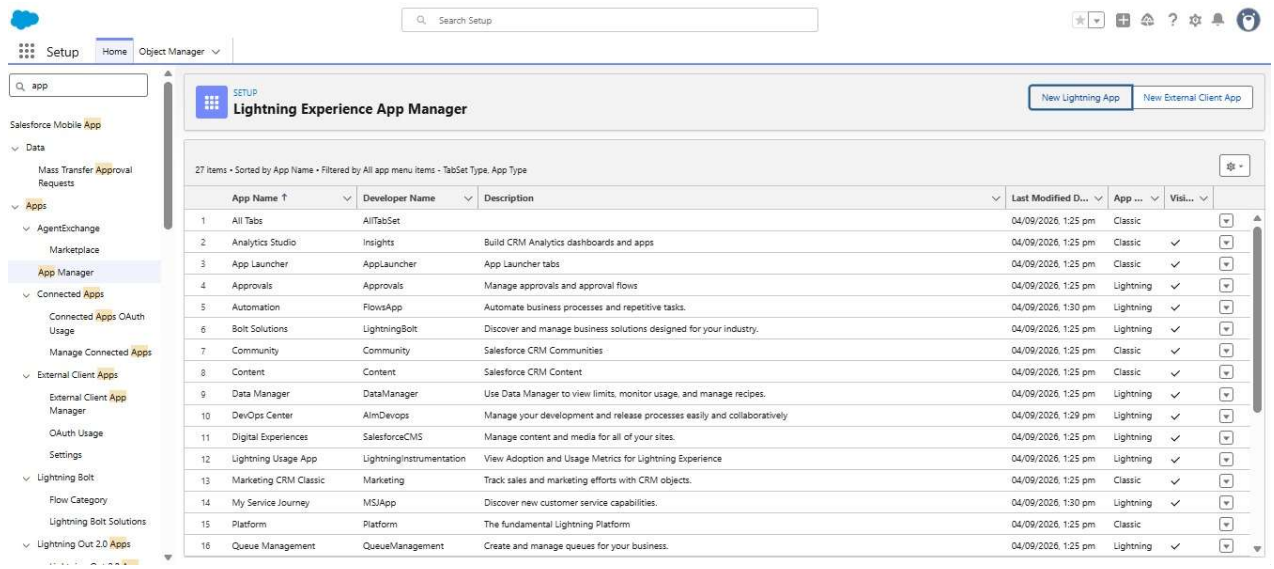
Milestone 4 - Lightning App:

What is an App?

Apps in Salesforce area group of tabs that help the application function by working together asa unit. It has a name,a logo, and a particular set of tabs. The simplest app usually has just two tabs.

Activity 1:

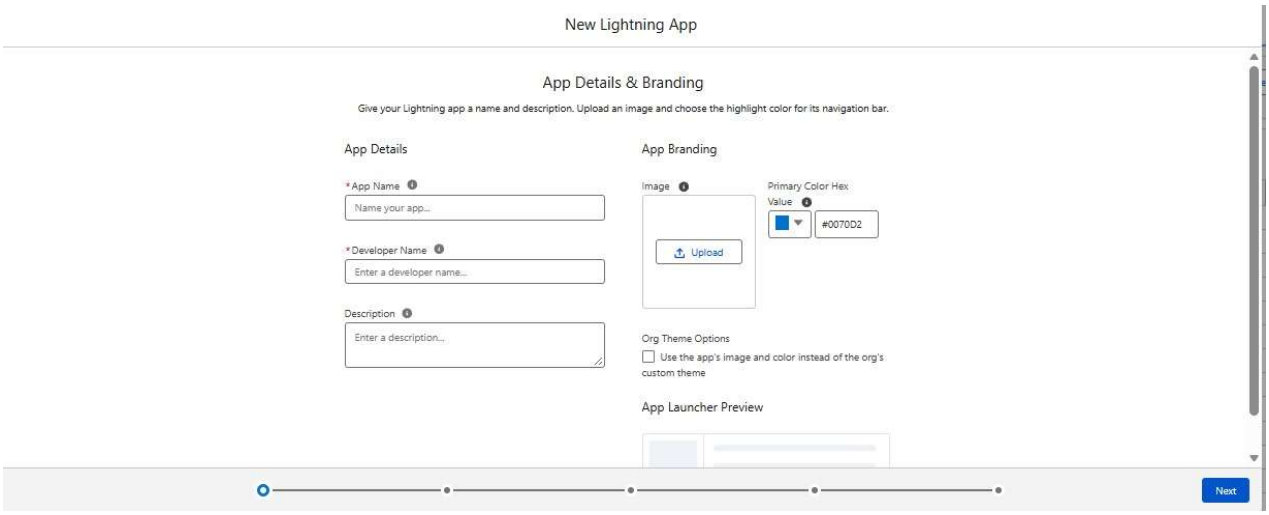
Create the Farm Management System app



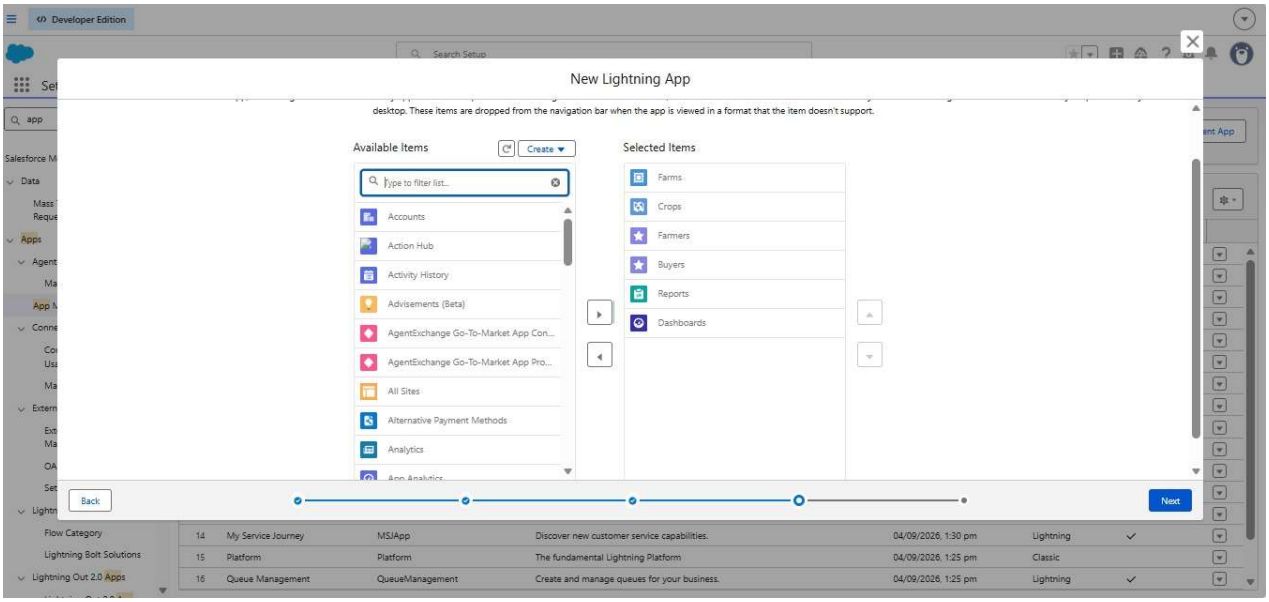
- From Setup, enter App Manager in the Quick Find and select App Manager.
- Click New Lightning App. Farm Management System as the App Name, then click Next
- Under App Options, leave the default selections and click Next.
- Under Utility Items, leave as is and click Next.
- From Available Items, select Farm, Crop,Farmer,Buyer Reports,and Dashboards and move them to Selected Items. Click Next.
- From Available Profiles, select System Administrator and move it to Selected Profiles Profiles Click Save & Finish.
- To verify your changes, click the App Launcher, type Farm Management System and select the Job Application app.

Note:

1. App Launcher-Displays available apps.
2. App Name-Displays the current selected app.



3.Navigation menu-Displays the tabs Available inside the app.



Milestone 5:

Fields and Relationship : What are fields?

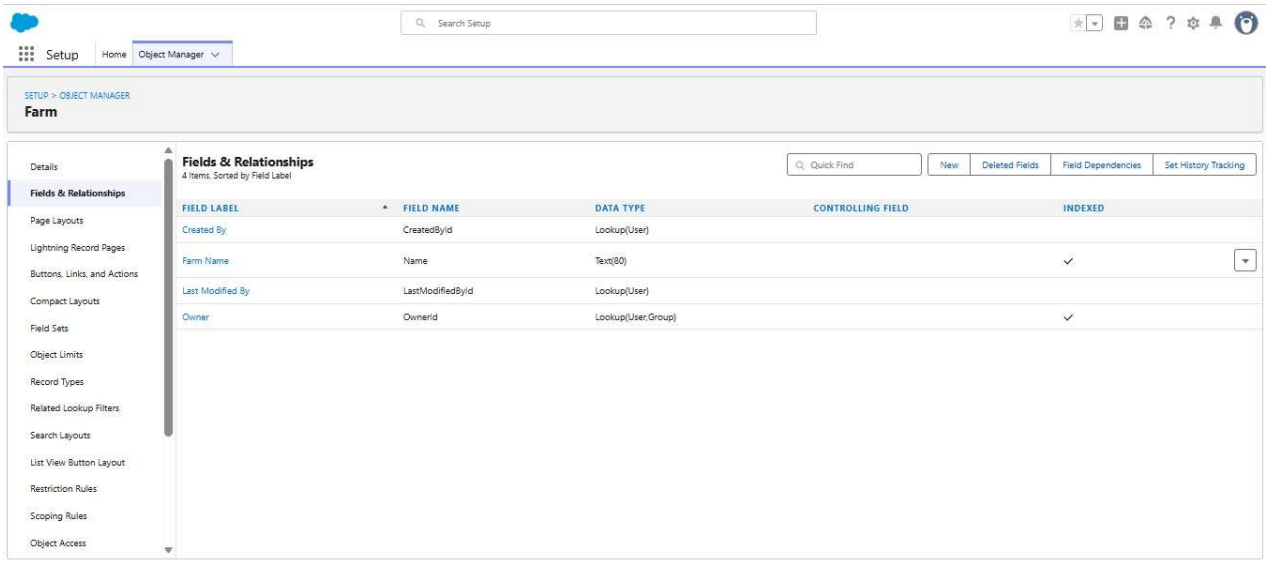
Fields in salesforce represent what the columns represent in relational databases. It can store data values which are required for a particular object in a record.

Activity 1:

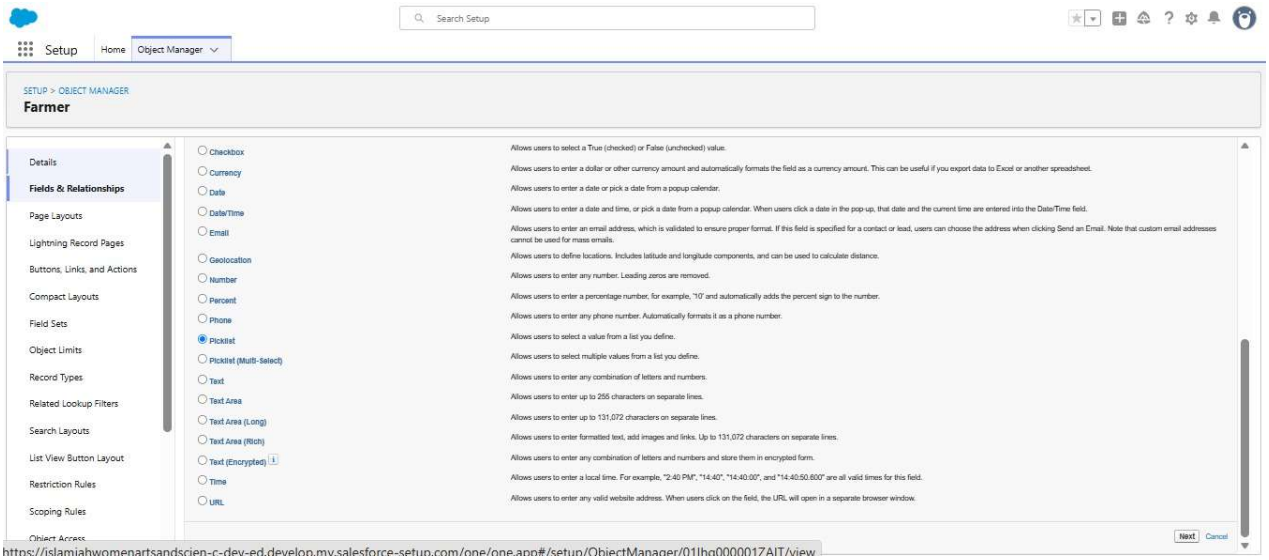
Creation of fields for the Farm objects:

1. click the gear icon and select Setup. This launches Setup in a new tab.
2. Click the Object Manager tab next to Home.

3. Select Farm.
4. Select Fields & Relationships from the left navigation, and click New.

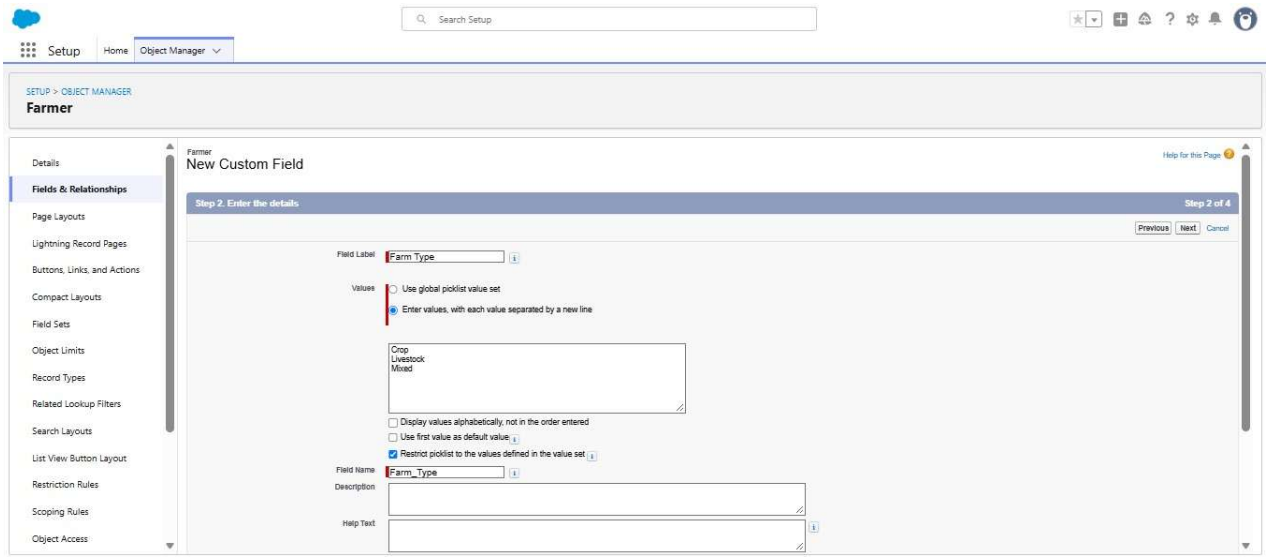


5 . Choose Data type as a picklist,Click next.



6. Enter Label(Farm Type), and Enter Values as (Crop Values as (Crop, Livestock,Mixed).

7. Next,Next Click & Save.



Now let's create the other fields following Activity1 steps 1 to 4 and we must choose the data types of the fields carefully.

1.Next we Create another Picklist Field,Enter field label (Soil Quality) enter values (Organic,Cover Crops,Composting)

2.Again we Create Picklist field,Field Label (Irrigation type), Enter Values (Sprinkler irrigation,Drip irrigation,localized irrigation)

3.Choose data type as a text,enter field label(Location)

4.Choose data type as text,enter field label (size).

5.Create a Master Detail relationShip with Farmer Object.

Activity 2:

Creation of fields for the Crop objects:

1.Choose Data type As a text,Enter Field Label name (Crop Name),Next save.

2.Select Data type as date ,Enter Field label (Planting Date),then save .

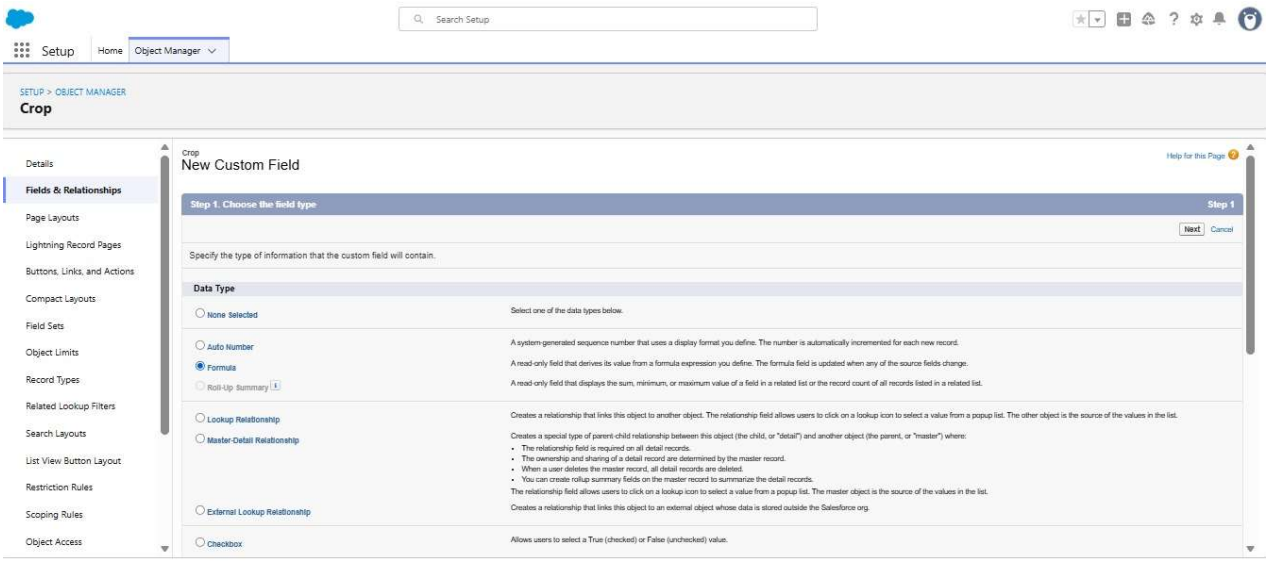
3.Again we select data type as a date,Enter field labe (Harvest date),save

4.Next We Create Lookup relationship on crop object realted to Farm Object.

5.We select adata type as currency,Enter field Label (Actual Yield), Next Save it.

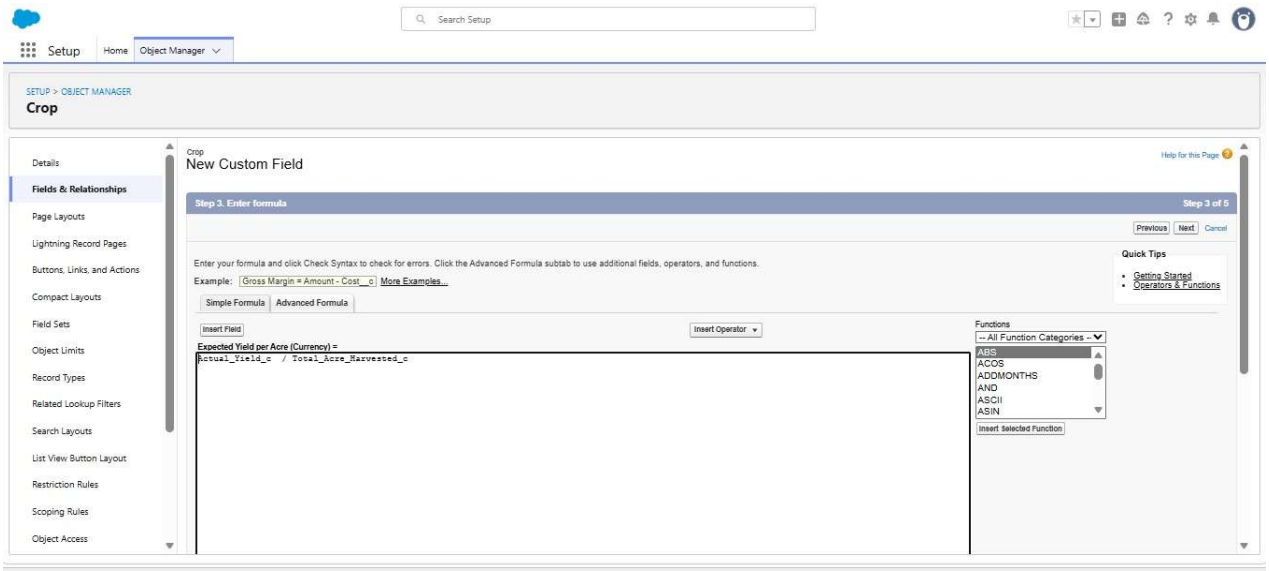
6.Select Data type as a Number,enter Field Label (Total Acres Harvested),Enter Maximum Length(20) and save.

7.Now We create Formula Field For That Select data Type As a Formula,Click Next.



8.Enter Field Label (Expected Yield per Acre),Select formula return type as Currency,click Next.

9.Enter a Formula Actual Yield Divide By Total Acre Harvested, then click next and Save.



Activity 3:

Creation of fields for the Farmer objects:

1.Select Data type Phone,Enter field label(Contact Number),next Save.

2.Select Data type Picklist,Enter Field Label(State),Enter values (Telangana,Andhra Pradesh,Tamilnadu),save.

3.Choose Data type Picklist,Enter Field Label (District),Enter Values of district,next save.

4.Again select Data type as Picklist,Enter Field Label (Village),Enter Values of villages,next save.

Creation of field dependency :

1.Now create Field Dependency For District field,Open Object Manager,Choose Farmer object ,select Fields&Relationships,On Right side We have Field Dependency option.

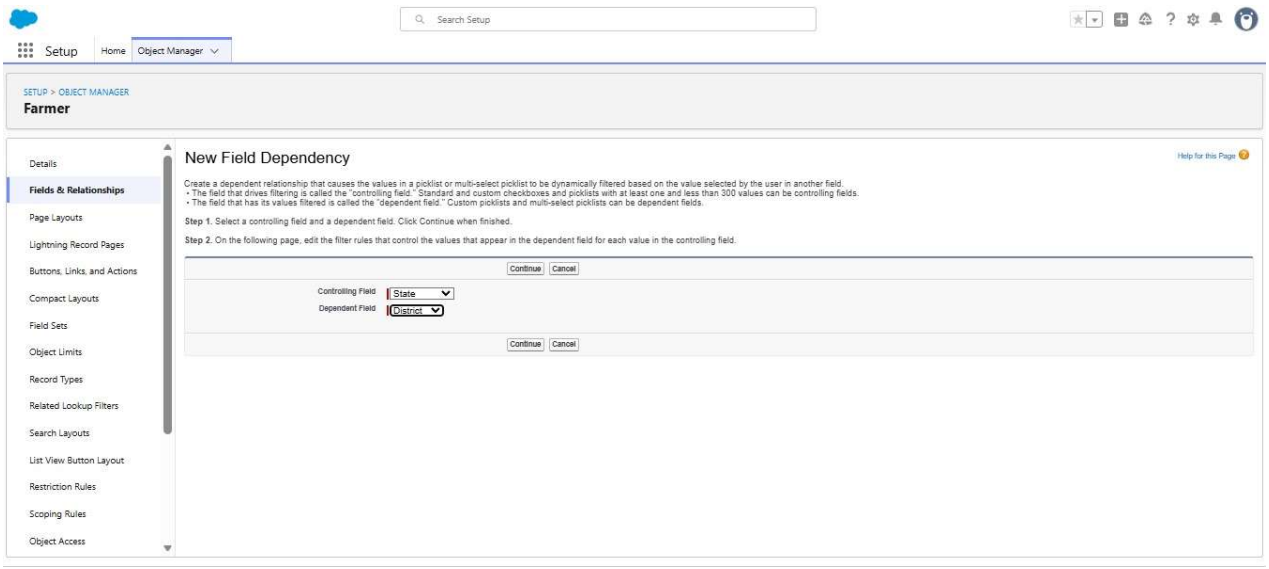
2.Click new, Select Controlling Field as State,Dependent Field As District,Click on Continue.

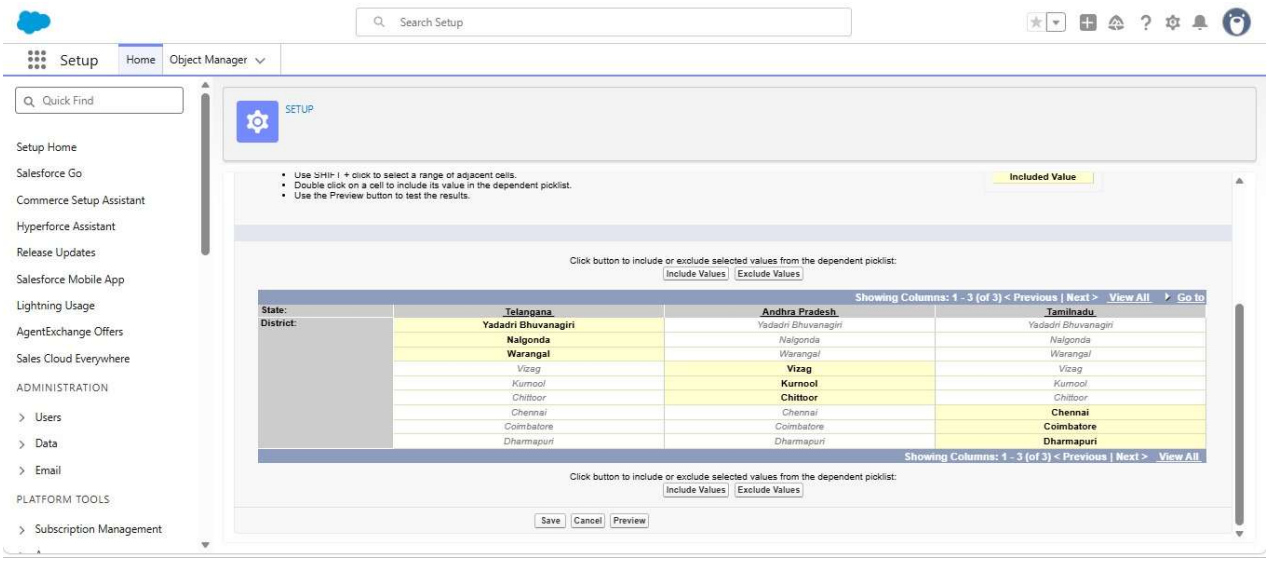
3. Select the appropriate Value in each column by double-clicking them.

4. For Ex. Telangana - Yadadri Bhongir.

5.Click Include Values. And it is also the same for Andhra Pradesh, Tamilnadu with its City.

6.Click Preview, then test the dependency by selecting a different State and viewing the associate Values available for Particular state.





7. Click, Close to close the preview window,Click Save.

Activity 4:

Creation of fields for the Buyer objects:

1.Select Data type as a Phone,Enter a Field Label(Contact),Save.
2.Select Data type as an Email,Enter Field Label(Email), Next save.
3.Select Data type Text area,Enter Field Label(Address),Save it.
4.Select Data type Picklist, Field Label(Buyer Type),Enter values (Wholesale,Retail),Click Save.

Milestone 6- Profile:

What is a profile?

A profile is a group/collection of settings and permissions that define what a user can do

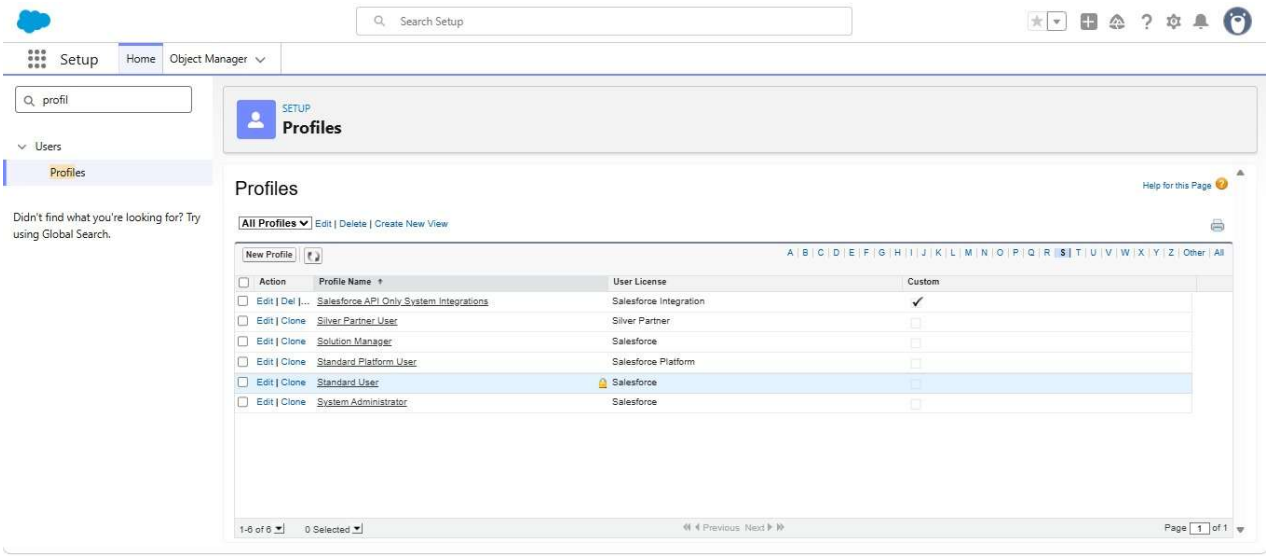
in salesforce. A profile controls “Object permissions, Field permissions, User permissions,Tab settings, App settings, Apex class access, Visualforce page access, Page layouts, Record Types, Login hours & Login IP ranges.

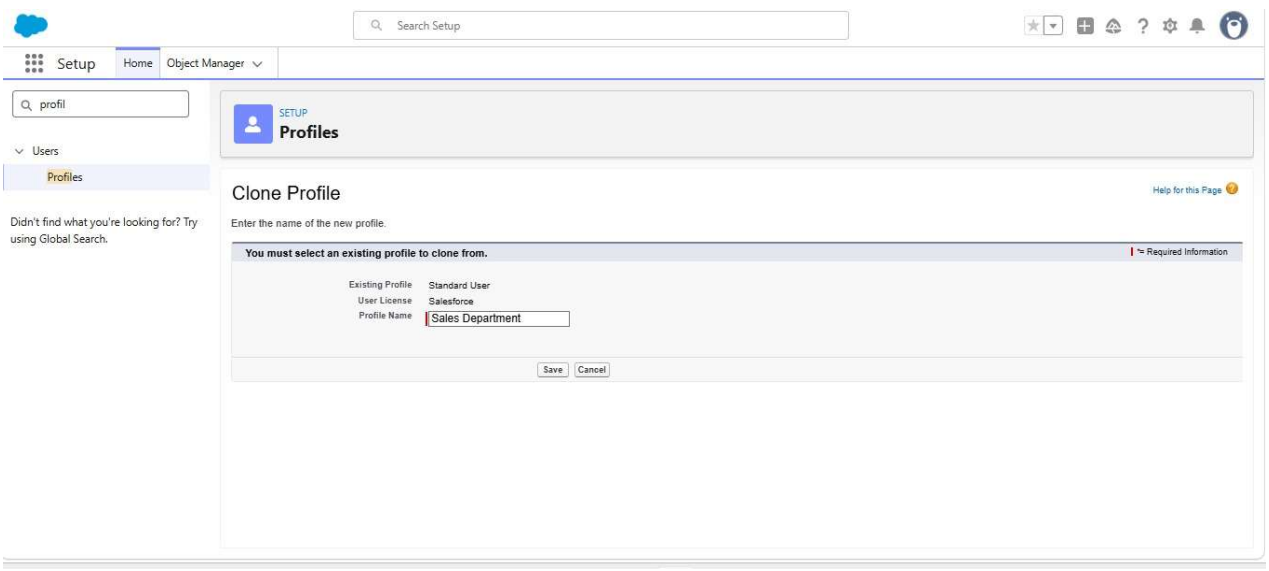
A profile can be assigned to many users, but user can be assigned single profile at a time.

Activity 1:

Create a custom profile:

1. From setup , enter profiles in Quick Find box
2. Select profiles (Standard user).
3. Click clone.
4. For Profile, Enter Sales Department.





5. Click save

Activity 2:

Create a custom profile-2:

Create a profile with the profile name as “ Farm Manager” .

6. From setup , enter profiles in Quick Find box
7. Select profiles (Standard user).
8. Click clone.

Milestone 7- Role

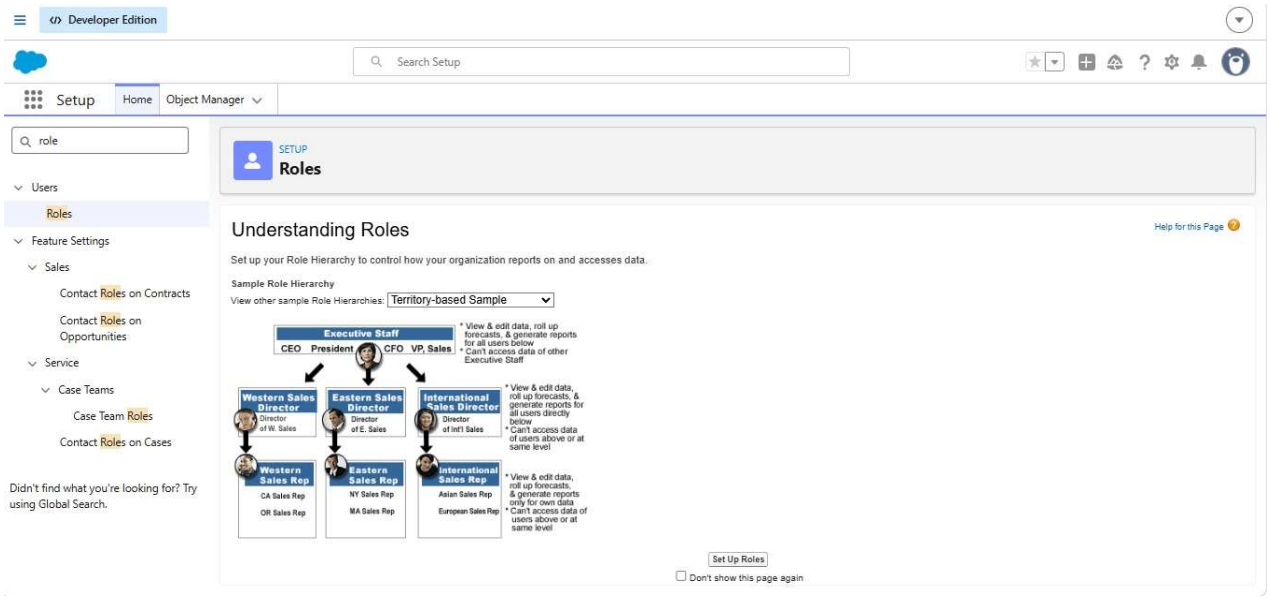
In Salesforce, roles are used to determine which users have access to certain data and

functions within the system. They are also used to define the reporting hierarchy within an organization. Users with higher roles have greater access to data and more control over the system.

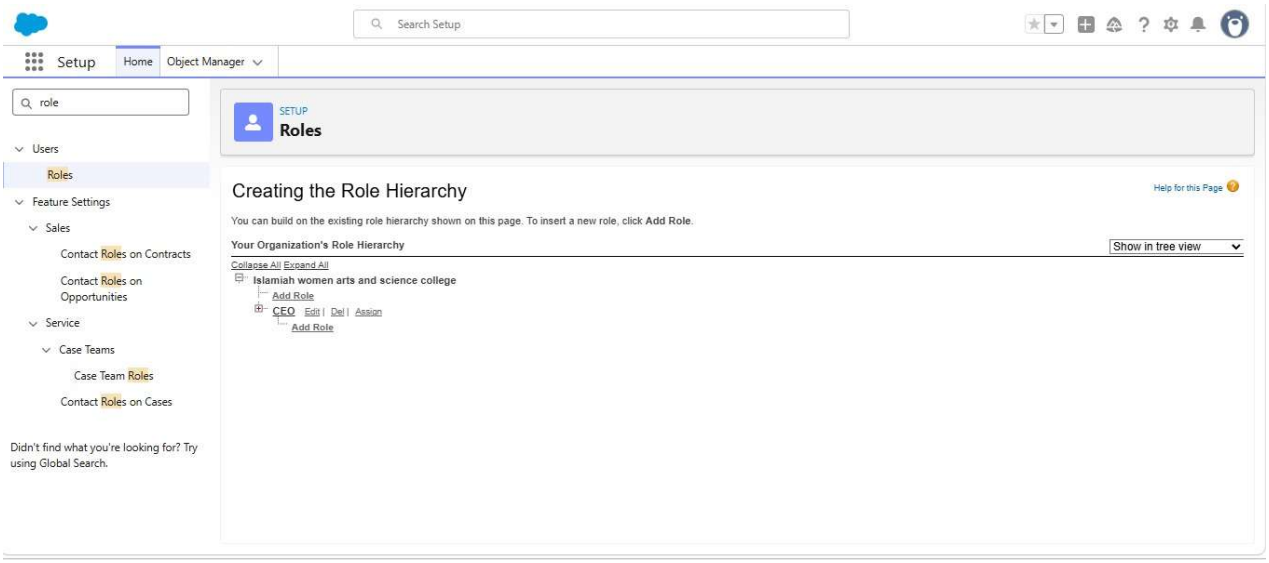
Activity-1

Creation of role

1. From the Quick find box search for the role and click on the roles option
2. select the set up roles option.



3. Below the CEO click on add role and enter the label name as a “Farm Manager” and role name will be Automatically populated and click on save.



Milestone 8-User :

What is a user?

A user is anyone who logs into Salesforce. Users are employees at your company, such as sales reps, managers, and IT specialists, who need access to the company's records.Every user in Salesforce has a user account. The user account identifies the user, and the user account settings determine what features and records the user can access.

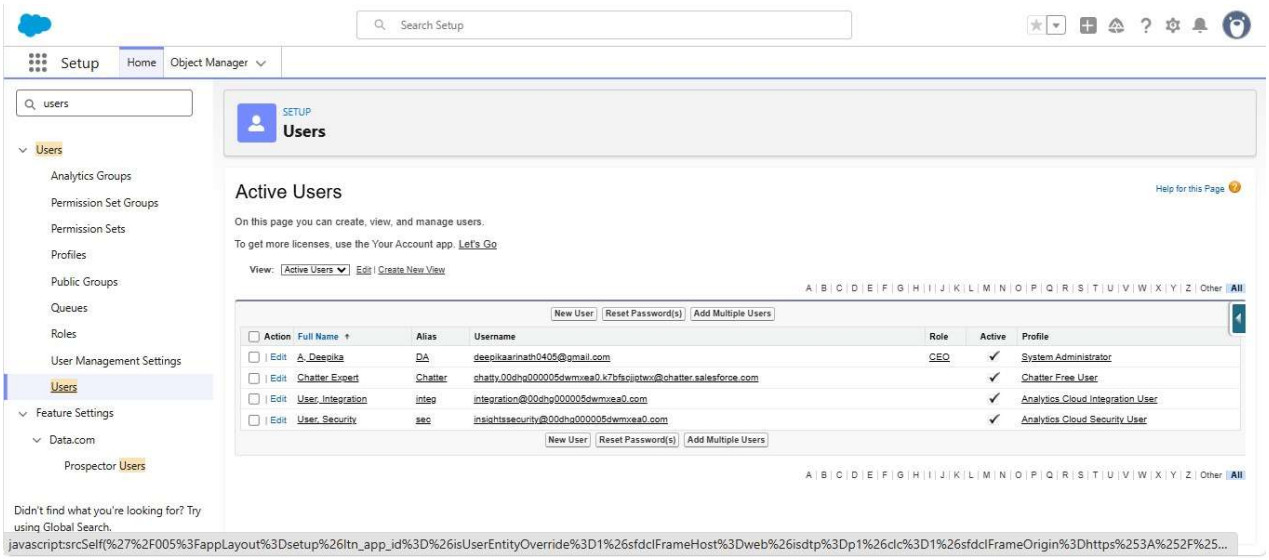
NOTE- As Salesforce license can only be used by 2 Users at a time in Dev Org, so If you don't find salesforce license then deactivate a user who has salesforce license Or change the license

type from Salesforce to any other.

Activity 1:

To Create a user:

1. From Setup, enter Users in the Quick Find box, then select Users.
2. Click New User.



3. Enter First name as Farm and last name as Manager.
4. Enter the user's name and email address and a unique username in the form of an email address. By default, the username is the same as the email address.
5. Select the role Farm Manager.
6. Select user License as Standard Platform User.
7. Select profile(Farm Manager).
8. Click save.

Activity 2:

Create a user with a username as "Sales Department", and assign him the sales Manager profile. Follow the steps from above Activity

Milestone 9-Sharing Rules:

What are Sharing Rules?

Sharing rules help users to share records based on conditions. It is basically created for objects whose organization-wide defaults (OWD) are set to public read-only or private because sharing rules can only extend the access and not restrict it.

Types of sharing rules,
•Owner-based Sharing Rules •Criteria-based Sharing Rules

Activity 1:

Create a sharing rule

1) Goto Sharing Settings, which can be found under the Quick Find section.

2) Scroll down and find the candidate object where a sharing rule needs to be added, and then click on New to create a new sharing rule.

3) Add the label of the sharing rule you want to make.

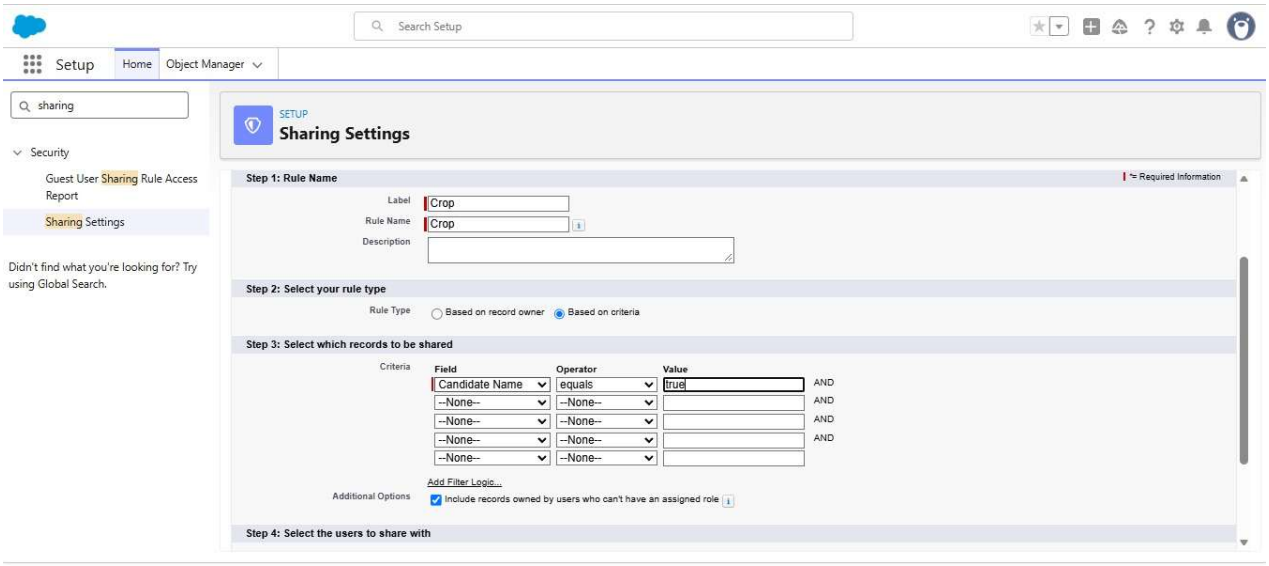
4) Select your rule type based on the criteria.

5) Select the field can join immediately check field from the Crop object.

6) Select the Crop as equal and value is True.

7) And in selecting the users to share with the section select roles and in that select Farm Manager.

8) And in the section of select the level of access for the users give the access Read/Write
.
9) And save the rule.



Activity 2:

Create a Sharing Rule to Share the records of Buyer to Sales department with the Access of Read/Write.

Follow the steps from above Activity.

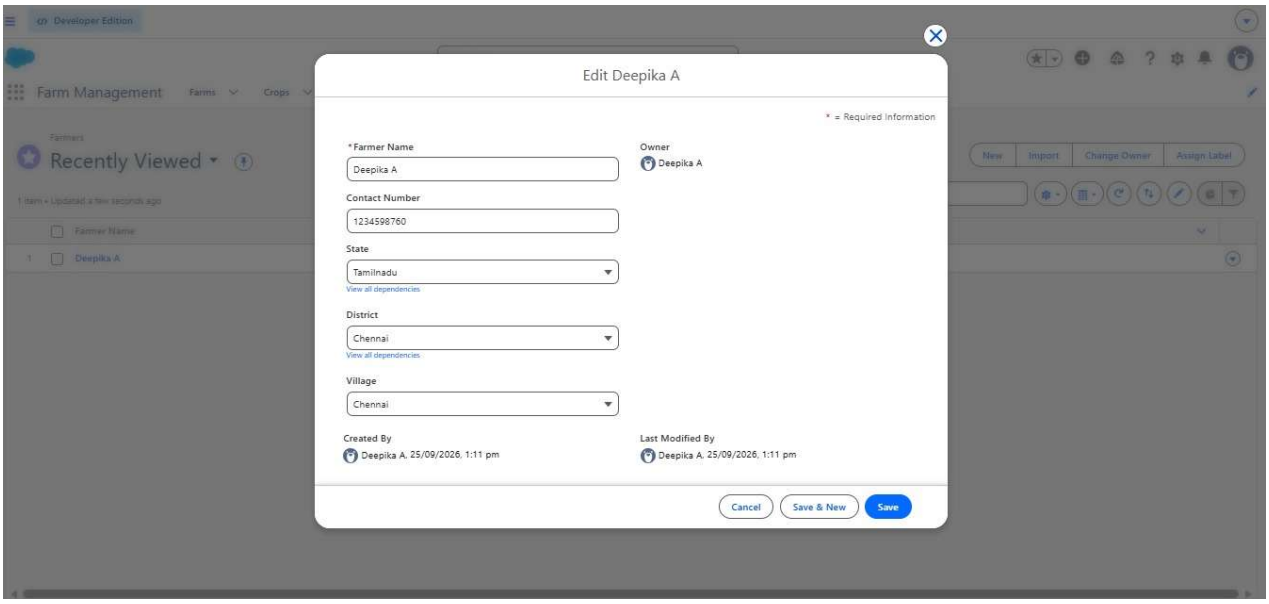
Milestone-10: User Adoption

Activity 1:Create a Record(Farmer)

- Click on App Launcher on left side of screen.
- Search Farm Management & click on it.

● Click on Farmer Tab.

● Click new and fill details & Save.



Milestone 11-Reports:

What are Reports?

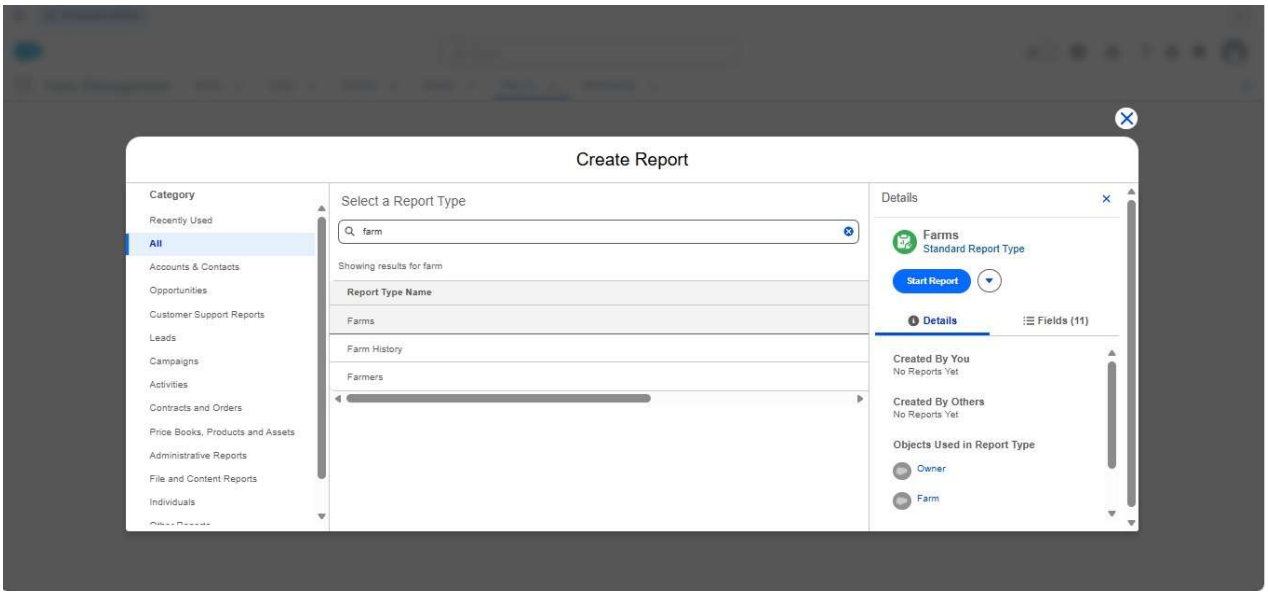
A report is a list of records that meet the criteria you define. It's displayed in rows and columns, and can be filtered, grouped, or displayed in a graphical chart. Every report is stored in a folder. Folders can be public, hidden, or shared, and can be set to read-only or read/write.

Activity 1:

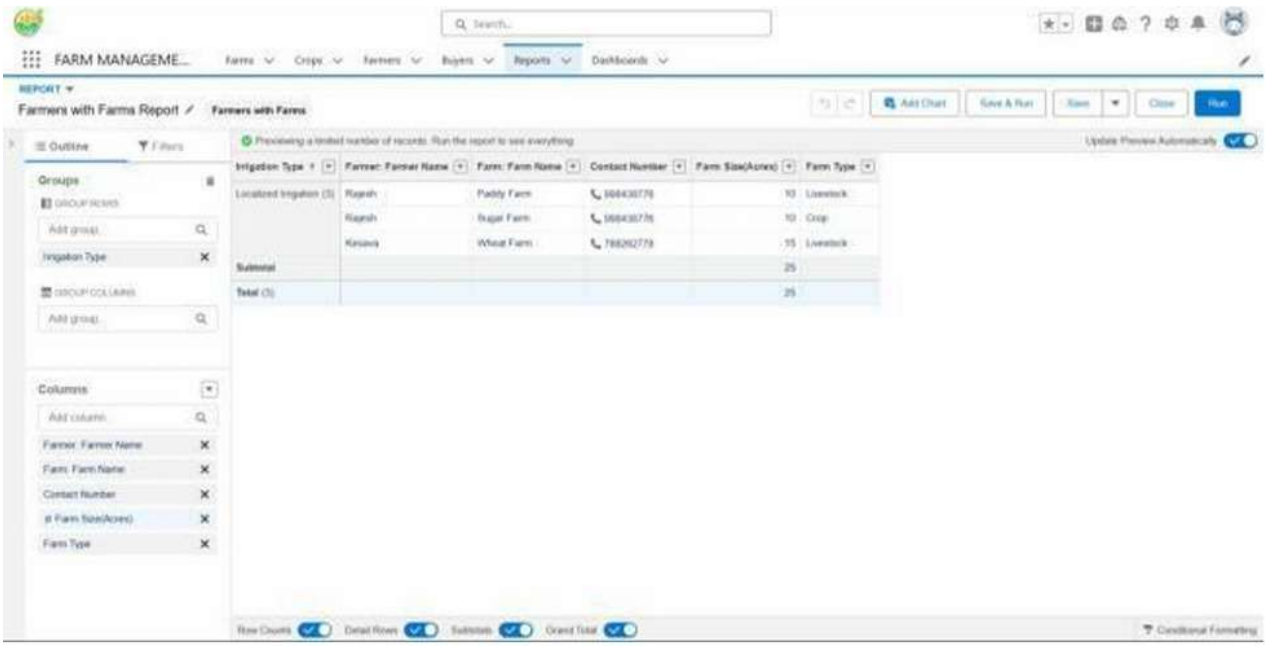
Create a report:

Create a report that displays rating of the account and which has type and account name.

1) Click on app launcher search for reports.
2) Click on the new report and select the category has Farmer with Farms.



3) In the details section select the option start report.
4) In Outline Click on Column Select Which Field you want to display in the Report.
5) Click Group,Select Which Fields You Want to Group.
6) Save the report by giving label name(Farmer with Farms) and save the folder as a public folder and save the report.



Activity 2:

Create a Report using the Objects Crops, Buyers. Follow the steps from above Activity.

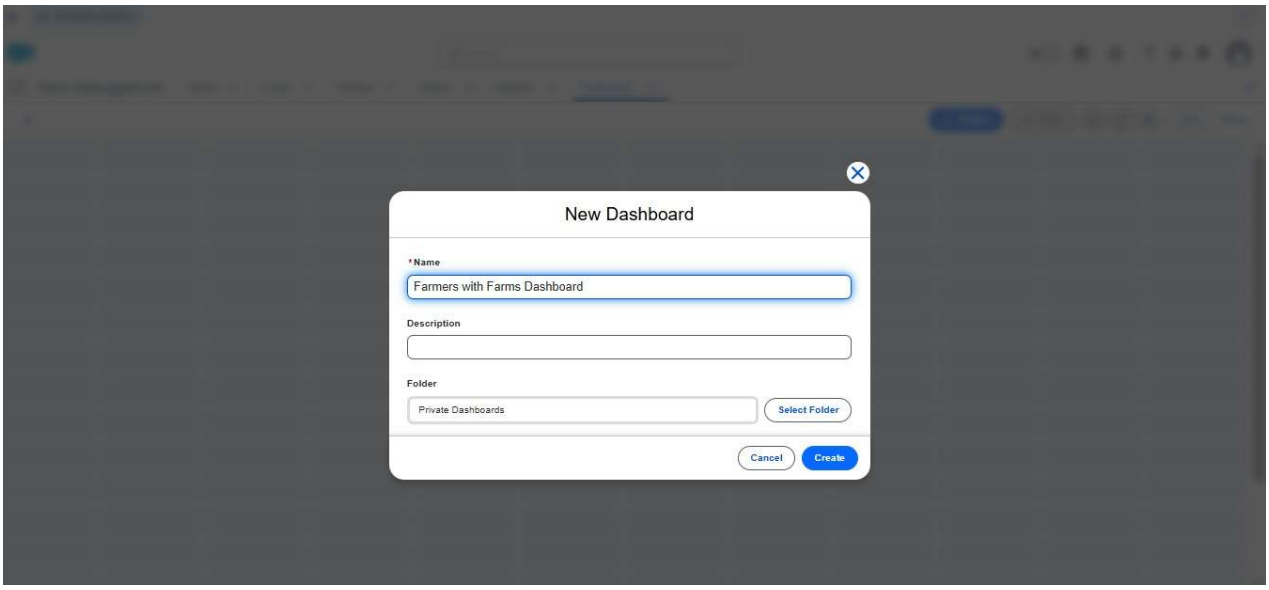
Milestone 12-Dashboard:

Dashboards provide more insights than reports as they combine the data from many reports and show a summarized result. Looking at many reports at a time gives the flexibility of combining the results from them quickly. Also summaries in dashboards help us decide on action plans quicker. The dashboards can contain charts, graphs and Tabular data.

Activity 1:

Create a Dashboard:

1. Click the Dashboard tab.
2. Click New Dashboard.
3. Name the dashboard Farmers with farms and click Create.
4. Click +Component.
5. Select the Farmers with Farms and click Select.
6. Select the Gauge chart and click Add.
7. Click Save and then Done.

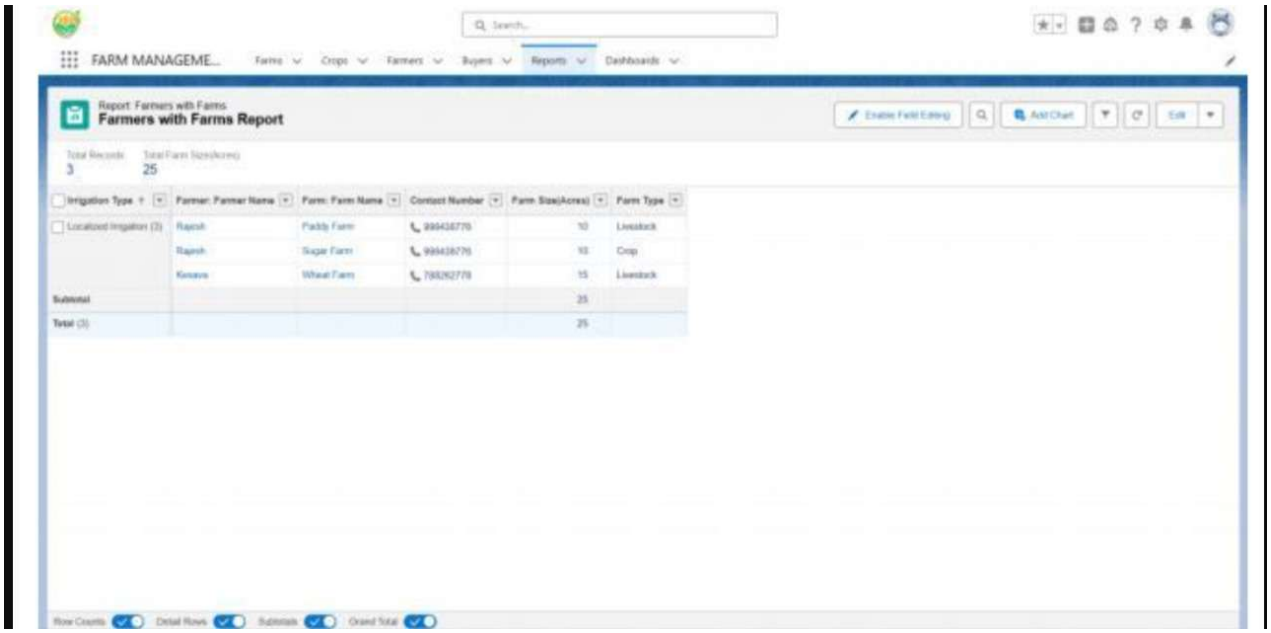


Milestone-13: View Report and Dashboard:

Activity 1: Report

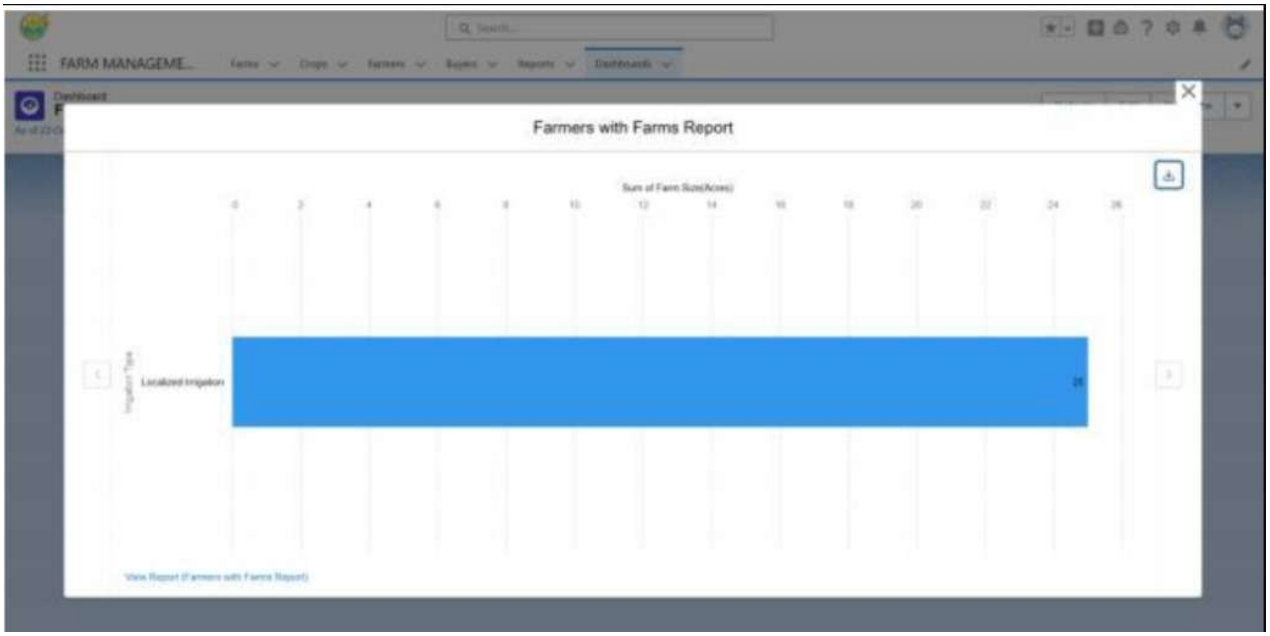
● Click on App Launcher on left side of screen.

- Search Farm Management System & click on it.
- Click on Reports Tab.
- Click on Farmers with Farms & see records



Activity 2: Dashboard

• Click on App Launcher on left side of screen.
• Search Farm Management system & click on it.
• Click on Dashboard Tab.
• Click Farmers With farms & see records.



7. FUNCTIONAL & PERFORMANCE TESTING

During the testing phase, screenshots were captured from the Salesforce interface to verify system functionality. The following were tested:

- Creation of Farm, Crop, Farmer, and Buyer records
- Validation of input fields
- Relationship and dependency checks
- Profile and role access testing
- Sharing rule verification
- Report and dashboard generation

Performance testing ensured smooth system response and stability during multiple user Operations.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized data system
- Reduced paperwork
- Improved accuracy
- Better monitoring
- Real-time analytics

Disadvantages

- Requires internet connection
- Salesforce license dependency
- Limited offline access

9. Conclusion:

The Farm Management System successfully digitizes agricultural data management using Salesforce. It improves efficiency, reduces errors, and enables farmers and managers to make informed decisions through automation and analytics.

10. FUTURE SCOPE

- Mobile application integration
- AI-based crop prediction
- Weather data integration

- IoT sensor connectivity
- Government scheme integration
- Market price forecasting

11. APPENDIX

- Source Code: Salesforce Configuration and Metadata
- Dataset: Farm, Crop, Farmer, and Buyer Records
- Tools Used: Salesforce Developer Org.

Thank you